P U R E M A T H E M A T I C S

Notes

1. In some areas, the Department of Pure Mathematics offers two distinct streams of courses, one for students in a pure mathematics major plan, and another for students in other majors. PMATH courses numbered from 345 to 352 are designed for pure mathematics majors; however, they are open to all students. The PMATH courses numbered from 331 to 336 cover similar topics at a less intensive level.
2. More detailed course descriptions and availability information can be obtained from the Pure Mathematics departmental web pages.

PMATH 300s

PMATH 320 LEC,TUT 0.50

Course ID: 015485

Euclidean Geometry

Euclid's axioms, triangle centres, conic sections, compass-and-straightedge constructions, isometries of the Euclidean plane and of Euclidean space, regular and star-shaped polygons, tessellations of the Euclidean plane, regular and quasi-regular polyhedra, symmetries of polygons and polyhedra, four-dimensional polytopes, sphere packings, and the kissing problem. Applications.

[Note: This course will be of interest to all math students.]

Prereq: (MATH 106 or 114 or 115 or 136 or 146 or 215 or NE 112) and (MATH 104 or 109 or 116 or 117 or 124 or 127 or 137 or 147)

PMATH 321 LEC 0.50

Course ID: 015486

Non-Euclidean Geometry

An introduction to three types of non-Euclidean geometry: spherical, projective and hyperbolic geometry. Lines, distances, circles, triangles, and areas in these non-Euclidean spaces. Conic sections in the projective plane. Inversions and orthogonal circles. Models of the hyperbolic plane (such as the Poincaré disc model or the upper-half plane model). Tilings of the hyperbolic plane.

[Note: This course will be of interest to all math students.]

Prereq: (MATH 106 or 114 or 115 or 136 or 146 or 215 or NE 112) and (MATH 104 or 109 or 116 or 117 or 124 or 127 or 137 or 147).

Antireq: PMATH 360.

PMATH 330 LEC,TST 0.50

Course ID: 007659

Introduction to Mathematical Logic

A broad introduction to mathematical logic. The notions of logical consequence and derivation are introduced in the settings of propositional and first order logic, with discussions of the completeness theorem and satisfiability.

[Note: PMATH 432 may be substituted for PMATH 330 whenever the latter is a requirement in an Honours plan.]

Prereq: (MATH 135 or 145) and (MATH 225 or 235 or 245); Not open to Computer Science students.

Antireq: CS 245, SE 212.

Also offered Online

PMATH 331 LEC 0.50

Course ID: 003323

Applied Real Analysis

Topology of Euclidean spaces, continuity, norms, completeness. Contraction mapping principle. Fourier series. Various applications, for example, to ordinary differential equations, optimization and numerical approximation.

[Note: PMATH 351 may be substituted for AMATH/PMATH 331 whenever the latter is a requirement in an Honours plan. Offered: F,W]

Prereq: MATH 237 or 247

(Cross-listed with AMATH 331)

PMATH 332 LEC,TST,TUT 0.50

Course ID: 003324

Applied Complex Analysis

Complex numbers, Cauchy-Riemann equations, analytic functions, conformal maps and applications to the solution of Laplace's equation, contour integrals, Cauchy integral formula, Taylor and Laurent expansions, residue calculus and applications.

[Note: PMATH 352 may be substituted for AMATH/PMATH 332 whenever the latter is a requirement in an Honours plan. Offered: W,S]

Prereq: MATH 237 or 247.

Antireq: PHYS 365

(Cross-listed with AMATH 332)

PMATH 333 LEC 0.50

Course ID: 015092

Introduction to Real Analysis

The purpose of the course is to present the familiar concepts of calculus at a rigorous level and to provide students who took the MATH 137/MATH 138/MATH 237 sequence with the background needed to be successful in PMATH 351 and PMATH 352. Topics discussed include the completeness properties of the reals; the density of the rationals; the topology of real n-dimensional space: open and closed sets, connectedness, compactness (by open covers), the Heine-Borel theorem, completeness; sequences in real n-dimensional space: convergence, Cauchy sequences, subsequences, the Bolzano-Weierstrass theorem; multivariable functions: limits, point-wise and uniform continuity, the extreme value theorem, uniform convergence of sequences of functions, Taylor's theorem, term-by-term differentiation of power series; integration in real n-dimensional space: Riemann integrability, Fubini's theorem for continuous functions on rectangles, term-by-term integration of power series.

Prereq: One of (MATH 128 with at least 70%), (MATH 138 with at least 60%), MATH 148.

Coreq: (MATH 235 or 245) and MATH 237.

Antireq: MATH 247

PMATH 334 LEC 0.50

Course ID: 007662

Introduction to Rings and Fields with Applications

Rings, ideals, factor rings, homomorphisms, finite and infinite fields, polynomials and roots, field extensions, algebraic numbers, and applications, for example, to Latin squares, finite geometries, geometrical constructions, error-correcting codes.

Prereq: MATH 235 or 245.

PMATH 336 LEC 0.50

Course ID: 007663

Introduction to Group Theory with Applications

Groups, permutation groups, subgroups, homomorphisms, symmetry groups in two and three dimensions, direct products, Polya-Burnside enumeration.

Prereq: MATH 235 or 245.

PMATH 340 LEC 0.50

Course ID: 007664

Elementary Number Theory

An elementary approach to the theory of numbers; the Euclidean algorithm, congruence equations, multiplicative functions, solutions to Diophantine equations, continued fractions, and rational approximations to real numbers.

[Note: PMATH 440 may be substituted for PMATH 340 whenever the latter is a requirement in an Honours plan.]

Prereq: MATH 225 or 135 or 145

PMATH 343 LEC 0.50

Course ID: 007665

Introduction to the Mathematics of Quantum Information

Finite dimensional normed vector spaces and inner product spaces. Positive and normal operators, the spectral theorem, and singular value decomposition. Tensor products, finite dimensional C* algebras, and the GNS representation. Completely positive maps, Stinespring's theorem, the Choi-Jamiolkowski isomorphism, and the Choi-Krauss representation. Entanglement and the Bell and Tsirelson inequalities. Vector states and density matrices, quantum channels, observables, and quantum measurement.

Prereq: (MATH 235 or 245) and (AMATH/PMATH 331 or MATH 247 or PMATH 333).

Antireq: PMATH 399 taken Winter 2019

PMATH 347 LEC 0.50

Course ID: 014182

Groups and Rings

Groups, subgroups, homomorphisms and quotient groups, isomorphism theorems, group actions, Cayley and Lagrange theorems, permutation groups and the fundamental theorem of finite abelian groups. Elementary properties of rings, subrings, ideals, homomorphisms and quotients, isomorphism theorems, polynomial rings, and unique factorization domains.

Prereq: MATH 235 or 245

PMATH 348 LEC 0.50

Course ID: 014183

Fields and Galois Theory

Fields, algebraic and transcendental extensions, minimal polynomials, Eisenstein's criterion, splitting fields, and the structure of finite fields. Sylow theorems and solvable groups. Galois theory. The insolvability of the quintic.

Prereq: PMATH 347

PMATH 351 LEC 0.50

Course ID: 007669

Real Analysis

Normed and metric spaces, open sets, continuous mappings, sequence and function spaces, completeness, contraction mappings, compactness of metric spaces, finite-dimensional normed spaces, Arzela-Ascoli theorem, existence of solutions of differential equations, Stone-Weierstrass theorem.

Prereq: MATH 247 or PMATH 333

PMATH 352 LEC 0.50

Course ID: 007672

Complex Analysis

Analytic functions, Cauchy-Riemann equations, Goursat's theorem, Cauchy's theorems, Morera's theorem, Liouville's theorem, maximum modulus principle, harmonic functions, Schwarz's lemma, isolated singularities, Laurent series, residue theorem.

Prereq: MATH 247 or PMATH 333

PMATH 365 LEC 0.50

Course ID: 003325

Differential Geometry

Submanifolds of Euclidean n-space; vector fields and differential forms; integration on submanifolds and Stokes's Theorem; metrics and geodesics; Gauss-Bonnet Theorem.

Prereq: (MATH 235 or 245) and (MATH 237 or 247)

PMATH 370 LEC 0.50

Course ID: 009496

Chaos and Fractals

The mathematics of iterated functions, properties of discrete dynamical systems, Mandelbrot and Julia sets.

[Note: Programming experience on one computer language with graphical output is recommended.]

Prereq: (One of MATH 118, 119, 128, 138, 148) and (One of MATH 114, 115, 136, 146, 225)

PMATH 399 RDG 0.50

Course ID: 007680

Readings in Pure Mathematics

Reading course as announced by the Department.

PMATH 400s

PMATH 432 LEC 0.50

Course ID: 007687

First Order Logic and Computability

The concepts of formal provability and logical consequence in first order logic are introduced, and their equivalence is proved in the soundness and completeness theorems. Goedel's incompleteness theorem is discussed, making use of the halting problem of computability theory. Relative computability and the Turing degrees are further studied.

Prereq: PMATH 347

PMATH 433 LEC 0.50

Course ID: 012623

Model Theory and Set Theory

Model theory: the semantics of first order logic including the compactness theorem and its consequences, elementary embeddings and equivalence, the theory of definable sets and types, quantifier elimination, and omega-stability. Set theory: well-orderings, ordinals, cardinals, Zermelo-Fraenkel axioms, axiom of choice, informal discussion of classes and independence results.

[Note: PMATH 348 is highly recommended.]

Prereq: PMATH 347

PMATH 440 LEC 0.50

Course ID: 007690

Analytic Number Theory

Summation methods, analytic theory of the Riemann zeta function, Prime Number Theorem, primitive roots, quadratic reciprocity. Dirichlet characters and infinitude of primes in arithmetic progressions, and assorted topics.

Prereq: PMATH 352

PMATH 441 LEC 0.50

Course ID: 007691

Algebraic Number Theory

An introduction to algebraic number theory; unique factorization, Dedekind domains, class numbers, Dirichlet's unit theorem, solutions of Diophantine equations.

Prereq: PMATH 348

PMATH 445 LEC 0.50

Course ID: 014184

Representations of Finite Groups

Basic definitions and examples: subrepresentations and irreducible representations, tensor products of representations. Character theory. Representations as modules over the group ring, Artin-Wedderburn structure theorem for semisimple rings. Induced representations, Frobenius reciprocity, Mackey's irreducibility criterion.

Prereq: PMATH 347

PMATH 446 LEC 0.50

Course ID: 014185

Introduction to Commutative Algebra

Module theory: classification of finitely generated modules over PIDs, exact sequences and tensor products, algebras, localisation, chain conditions. Primary decomposition, integral extensions, Noether's normalisation lemma, and Hilbert's Nullstellensatz.

Prereq: PMATH 347.

Coreq: PMATH 348.

PMATH 450 LEC 0.50

Course ID: 007674

Lebesgue Integration and Fourier Analysis

Lebesgue measure on the line, the Lebesgue integral, monotone and dominated convergence theorems, Lp-spaces: completeness and dense subspaces. Separable Hilbert space, orthonormal bases. Fourier analysis on the circle, Dirichlet kernel, Riemann-Lebesgue lemma, Fejer's theorem, and convergence of Fourier series.

Prereq: PMATH 351 with a grade of at least of 60%

PMATH 451 LEC 0.50

Course ID: 003348

Measure and Integration

General measures, measurability, Caratheodory Extension theorem and construction of measures, integration theory, convergence theorems, Lp-spaces, absolute continuity, differentiation of monotone functions, Radon-Nikodym theorem, product measures, Fubini's theorem, signed measures, Urysohn's lemma, Riesz Representation theorems for classical Banach spaces.

Prereq: PMATH 450 with a grade of at least 60%

Also offered Online

PMATH 453 LEC 0.50

Course ID: 003349

Functional Analysis

Banach and Hilbert spaces, bounded linear maps, Hahn-Banach theorem, open mapping theorem, closed graph theorem, topologies, nets, Hausdorff spaces, Tietze extension theorem, dual spaces, weak topologies, Tychonoff's theorem, Banach-Alaoglu theorem, reflexive spaces.

Prereq: PMATH 450

PMATH 464 LEC 0.50

Course ID: 010733

Introduction to Algebraic Geometry

An introduction to algebraic geometry through the theory of algebraic curves. General algebraic geometry: affine and projective algebraic sets, Hilbert's Nullstellensatz, co-ordinate rings, polynomial maps, rational functions and local rings. Algebraic curves: affine and projective plane curves, tangency and multiplicity, intersection numbers, Bezout's theorem and divisor class groups.

Prereq: PMATH 347.

Coreq: PMATH 348.

PMATH 465 LEC 0.50

Course ID: 003350

Smooth Manifolds

Point-set topology; smooth manifolds, smooth maps, and tangent vectors; the tangent and cotangent bundles; vector fields, tensor fields, and differential forms; Stokes's theorem; integral curves, Lie derivatives, the Frobenius theorem; de Rham cohomology.

Prereq: PMATH 365

PMATH 467 LEC 0.50

Course ID: 007704

Algebraic Topology

Topological spaces and topological manifolds; quotient spaces; cut and paste constructions; classification of two-dimensional manifolds; fundamental group; homology groups. Additional topics may include: covering spaces; homotopy theory; selected applications to knots and combinatorial group theory.

Prereq: PMATH 347, 351.

PMATH 499 RDG 0.50

Course ID: 007706

Readings in Pure Mathematics

Reading course as announced by the Department.